ANSWERS

1.

Question	Answer	Marks
1(a)(i)	particles move / collide / hit	**B1**
	particles collide with (inner) walls (of rocket)	**B1**
	force exerted by collisions over / on an area	**B1**
1(a)(ii)	more collisions per unit time	**B1**
1(b)	(expelled) water gains / has (downward) momentum	**B1**
	momentum conserved	**B1**
	rocket gains (upward) momentum (and accelerates)	**B1**
1(c)	any **two** from: mass (of water and rocket) decreases / changes pressure (of air) decreases (air) resistance (acts) gravitational field strength decreases (with height) less water expelled per second / water expelled at slower speed	**B2**

2.

Question	Answer	Marks
2(a)	(p =) mv **or** 300×8000	**C1**
	2.4×10^6 (kg m / s)	**A1**
2(b)(i)	momentum conserved **or** $mv = m_1v_1 + m_2v_2$	**C1**
	(v_2 =) $(2.4 \times 10^6 - (150 \times 9000)) \div 150$ **or** (v_2 =) (1 050 000) / 150	**C1**
	7000 (m / s)	**A1**
2(b)(ii)	from energy stored as chemical (potential) energy	**B1**
2(b)(iii)	Δp **or** J **or** $I = 150 \times (9000 - 8000)$ **or** 150 000 **or** (a =) $(9000 - 8000) \div 0.20$ **or** (a =) 5000	**C1**
	(F =) $\Delta p / (\Delta)t$ **or** (F =) $I / (\Delta)t$ **or** (F =) ma **or** $150 \times (9000 - 8000) / 0.20$ **or** $150 \times 1000 / 0.20$	**C1**
	7.5×10^5 (N)	**A1**

3.

Question	Answer	Marks
2(a)(i)	the product of mass and velocity	**B1**
2(a)(ii)	kg m / s **or** Ns	**B1**
2(b)	$1100 \times 10 = 4100 \times v$	**C1**
	2.7 (m / s)	**A1**
2(c)(i)	(KE =) $\frac{1}{2} mv^2$ algebraic or numerical for the car	**C1**
	15 000 (J)	**A1**
2(c)(ii)	(kinetic energy) becomes heat / thermal energy / internal energy (of vehicles / air) **or** kinetic energy (of car) goes to kinetic energy **of van**	**B1**

4.

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	**B1**
	deceleration • has (average) value 0.39 m / s^2 • lasts for 40 s • is non-uniform	**B1**
1(b)(i)	friction (with road) **and** <u>air</u> resistance **or** (air) drag	**B1**
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) **or** (graph shows) deceleration is not constant / decreases **or** non-uniform decrease in speed	**B1**
1(c)(i)	(momentum =) mv in any form algebraic or numerical	**C1**
	8400 **and** kg m / s **or** Ns	**A1**
1(c)(ii)	(F =) $\Delta mv / t$ **or** $F = ma$ in any form algebraic or numerical	**C1**
	840 (N)	**A1**